

Julius Schneider

Ekkehardstrasse 5, 8006 Zurich (CH) • +41 (0) 78 245 45 65 • siulius@ethz.ch
<https://www.linkedin.com/in/julius-schneider-813696216/> • <https://www.julius-schneider.com>

EDUCATION

ETH Zurich (Bachelor in Computer Science) Focus on Computer Vision, ML and Autonomous Systems	Zurich (CH) 09/2023 – 09/2027
Purdue University (Exchange Semester) Joined PART (https://www.purdueaerial.com) and led field-testing and Computer Vision development	West Lafayette (USA) 08/2026 – 01/2027
Kantonsschule Kreuzlingen (Swiss Matura) (Swiss High School, English speaking class) GPA: 5.4/6; Core subjects Economics and Computer Science. Matura Paper: 6.0/6	Kreuzlingen (CH) 06/2019 – 06/2023

WORK EXPERIENCE

Computer Vision Engineer CDDS (https://www.cdds.ai) Worked on Pose Estimation Models and End-to-End Learning approaches for flight controls <i>Part-time Role</i> - Worked on expanding the Computer Vision pipeline with a pose estimation model for final trajectory predictions optimized for low-latency inference on edge hardware (NVIDIA Jetson Orin Nano) - Engineered synthetic data generation pipelines in NVIDIA Isaac to train models for scenarios where real-world data is dangerous to collect - Transferred Simulation from Gazebo to NVIDIA Isaac to enable end-to-end learning approaches - Led cooperation with CIHBw (Germany), DARPA, DIU (USA) and Taskforce Drones (Switzerland) during field testing	Zurich (CH) 02/2026 – 08/2026
Teaching Assistant ETH Zurich Teaching Assistant in the Courses Introduction to Prog. & Digital Design and Computer Architecture <i>Part-time Role</i> - Prepared and presented weekly classes to >30 Students - Reviewed and corrected the official final exam	Zurich (CH) 02/2025 – 06/2026
Falke Digital (Drone photography and image videos of Lake Constance region, www.falkedigital.ch) <i>Founder</i> - Built and operated high-performance FPV drones in cluttered environments to capture dynamic footage - Directed and produced professional image films for customers like Mercedes-Benz Switzerland - Achieved a substantial growth in revenue: 2022 turnover: CHF 17k; 2023 turnover: CHF 41k	Kreuzlingen (CH) 11/2018 – Present

PUBLICATIONS AND SELECTED PROJECTS

MIVE: Markerless Inertial Vector Extraction for Monocular Quadrotor Pose Estimation (Paper) Authored a paper about the developed data-generation pipeline and evaluated against conventional approaches <i>Part-time Role</i> - Developed a novel pipeline to generate ground truth data for thrust vector estimation in UAVs - Evaluated our approach against both synthetic and ArUco based data gathering approaches	Zurich (CH) 02/2026 – 07/2026
Second Place at European Defense Tech Hackathon Built an autonomous low-cost Drone Detection System using Microphones and Cameras <i>Part-time Role</i> - Sensor activated by the sound of an incoming drone, started searching and classifying the UAV - Sound detection via spectrogram analysis using a CNN and Drone classification with a finetuned YOLO model - Worked on the software and pitched the concept to a Jury of Investors and Defense Representatives	Zurich (CH) 11/2025
Third Place at the ETH PhysicalAI Hackathon organized by Helbling Built a pipeline that estimates the traveled distance of a sewer cleaning robot solely based on video data <i>Part-time Role</i> - After preprocessing we used an Optical Flow algorithm with Heuristics to judge the traveled distance - Additionally trained a CNN to flag frames in which cracks or obstructions were observable in the pipes	Zurich (CH) 05/2026
Semester Spokesperson for ETH Computer Science Student representative in Digital Design and Computer Architecture <i>Part-time Role</i> - Represented the interests of >500 Students in front of the board of Professors - Facilitated regular meetings with faculty and administration to address students' concerns	Zurich (CH) 04/2024 – 09/2024

SKILLS, INTERESTS

Languages	German native, English fluent, French fluent, Russian intermediate
AI & Robotics	PyTorch, OpenCV, NVIDIA Isaac Sim, C++, C, Gazebo, PX4, Linux (Ubuntu), Git, Docker Various Projects in the areas of Drone detection, and end-to-end learning for control Theory. Matura Paper: Development of a Python-based trading algorithm which performs fully automated buying and selling decisions for stocks. Backtested against SMI and other benchmarks. Grade: 6.0/6.0
Skills	Forward Deployed Engineering, Rapid development of prototypes, Licensed UAV Pilot,